Yuzong Chen

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EDUCATION

- **Cornell University** *Aug. 2022 - Present*
  Ph.D. in Electrical and Computer Engineering
  Advisor: Prof. Mohamed Abdelfattah
- **Nanyang Technological University, Singapore** *Aug. 2015 - Jun. 2019*
  B.Eng. in Electrical & Electronic Engineering
  GPA: 4.74 / 5.00, Honours (Highest Distinction)

RESEARCH INTERESTS

Efficient Hardware Architectures for Deep Learning: I am interested in developing new hardware for accelerating deep neural networks, with a special focus on sparsity and mixed precision. This includes deep neural networks compression and quantization, FPGA/ASIC design, and emerging hardware architectures such as computing in-memory.

EXPERIENCE

- **Graduate Research Assistant,** Cornell University *Aug. 2022 - Present*
  Advisor: Prof. Mohamed Abdelfattah
  Currently working on efficient hardware accelerators for sparse and mixed-precision deep neural networks.
- ***Research Engineer*,** National University of Singapore *Sept. 2021 – Jul. 2022*
  Advisor: Prof. Heng Chun-Huat
  Work on a joint project with NXP Semiconductors to design an RF switched-capacitor power amplifier for high-speed communication. Help tape-out the chip in 22nm FDSOI technology.
- ***Project Officer,*** Nanyang Technological University *Feb. 2020 - Aug. 2021*
  Advisor: Prof. Tony Tae-Hyoung Kim
  Conduct and lead projects about computing in-memory circuit design based on static random access memory (SRAM) and resistive random access memory (ReRAM). Help tape-out several chips in 65nm technology.
- **Deep Learning Intern,** Seagate Technology *May. 2018 – Aug.2018*
  Work on high-resolution image classification to detect the defects on hard-disk.
- **Control Systems Intern,** ExxonMobil Chemical Operations *Aug. 2017 – Dec. 2017*
  Set up industrial servers to facilitate data migration and maintenance. Help improve the automation software.

Selected PUBLICATIONS

- **Journal Articles**

[1] Yuzong Chen, Junjie Mu, Hyunjoon Kim, Lu Lu, and Tony Tae-Hyoung Kim, " BP-SCIM: A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory," *IEEE TCAS-I*, 2023.

[2] Yuzong Chen, Lu Lu, Bongjin Kim, and Tony Tae-Hyoung Kim, "A Reconfigurable 4T2R ReRAM Computing In-Memory Macro for Efficient Edge Applications," *IEEE OJCAS*, 2021.

[3] Yuzong Chen, Lu Lu, Bongjin Kim, and Tony Tae-Hyoung Kim, "Reconfigurable 2T2R ReRAM Architecture for Versatile Data Storage and Computing In-Memory," *IEEE TVLSI*, 2020.

- **Conference Proceedings**

[1] Yuzong Chen, Jordan Dotzel and Mohamed S. Abdelfattah, "M4BRAM: Mixed-Precision Matrix-Matrix Multiplication in FPGA Block RAMs," *IEEE FPT*, 2023.

[2] Yuzong Chen and Mohamed S. Abdelfattah, "BRAMAC: Compute-in-BRAM Architectures for Multiply-Accumulate on FPGAs," *IEEE FCCM*, 2023.

[3] Yuzong Chen, Junjie Mu, Hyunjoon Kim, Lu Lu, and Tony Tae-Hyoung Kim, "A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory," *IEEE ISCAS*, 2022.

[4] Yuncheng Lu, Zehao Li, Yuzong Chen, and Tony Tae-Hyoung Kim, "A 181μW Real-Time 3-D Hand Gesture Recognition System based on Bi-directional Convolution and Memoryless Clustering," *IEEE CICC*, 2022.

[5] Yuzong Chen, Lu Lu, Yuncheng Lu, and Tony Tae-Hyoung Kim, "A Multi-Functional 4T2R ReRAM Macro Enabling 2-Dimensional Access and Computing In-Memory," *IEEE ISCAS*, 2021.

HONOURS AND AWARDS

- Qualcomm Innovation Fellowship (QIF) Finalist, 2024
- Cornell Graduate Fellowship, 2022 - 2023
- Undergraduate Dean's List (top 5% of the cohort), academic year 2015, 2016
- Singapore Science and Engineering Undergraduate Scholarship for 4 academic years

TECHNICAL SKILLS

- Python, C++, SystemVerilog, Pytorch, Git, Cadence Virtuoso, Vivado HLS, Synopsys Design Compiler.

TEACHING EXPERIENCE

- CS 5785 (Applied Machine Learning), Fall 2023

EDITORIAL SERVICE

- **Journal Reviewer:** IEEE Transactions on Very Large Scale Integration Systems (TVLSI); IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I); IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS). IEEE Open Journal of Circuits and Systems (OJCAS).
- **Conference Secondary Reviewer:** International Symposium on Field-Programmable Gate Arrays (FPGA), Design Automation Conference (DAC).